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Siamese network-based user-independent model for surface electromyogram biometric authentication

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Abstract
The advent of deep-learning technology has enhanced the performance of biometric systems, including facial and fingerprint recognition systems. Although facial recognition is now commonly used for authenticating mobile device users, it can be easily falsified as it relies on external traits. In this study, we design a user-independent model for surface electromyogram (sEMG) biometric authentication using the convolutional Siamese network with an *N*-pair loss function. We then implement the shift-equivariant model by exploiting the convolution and padding operations to deal with small shifts in multichannel sEMG sensors for various users. Additionally, the augmentation methods for time-series data and spectrograms are used to further improve the performances of the model. We employ the public Gesture Recognition and Biometrics electroMyogram (GRABMyo) dataset, comprising 43 subjects and 16 gestures collected over 3 days, to train and evaluate the model. The proposed model achieves equal error rates of 5.62% and 8.94% for unknown subjects while preserving and leaking the gesture code, respectively. In cross-day experiments, the model achieves rates of 4.92% and 7.56%, respectively, demonstrating robustness to intersession variations.

KEYWORDS
biometrics, deep learning, Siamese network, surface electromyogram, user authentication

1 | INTRODUCTION

Biometric authentication systems leverage the unique physiological and behavioral characteristics of individuals to verify their identities. Face- and fingerprint-based biometric systems are widely deployed in mobile devices—such as smartphones and laptops—owing to their high accuracy and low sensor cost. However, these biometric traits must be revealed during the recognition process to enable their capture and subsequent synthetic generation using three-dimensional (3D) printers, lenses, and silicon [1]. Electrical biosignals—such as electrocardiograms (ECGs), electroencephalograms (EEGs), and electromyograms (EMGs)—have been widely investigated [1–7] because of their inherent property, namely, liveness, which can complicate their capture and synthesis. In particular, surface EMGs (sEMGs)—which noninvasively capture biological signals from the skin to measure the electrical currents associated with muscle activities—have been studied over the past decade in the field of biometrics [1, 8–18]. Although sEMG-based biometric systems have attracted less attention compared with their ECG- or EEG-based counterparts, they exhibit two advantages over these systems—that is, acceptable accuracy and easy customization of the registered template if compromised [18]. Moreover, the emergence of wearable devices—such as the meta wristband [19] and OptiTrack gloves [20]—that can interact with virtual or augmented reality has further increased the relevance of sEMG-based biometric systems.

Hand-gesture recognition (HGR) using sEMG signals [21–23] has been investigated because these signals are produced by muscle activities during hand movements. However, an HGR classifier should be calibrated or retrained for each individual to manage the influence of various physiological factors on sEMG signals, including the muscle fiber patterns, muscle blood flow changes, neural activities, neurotransmitter activities in different muscle regions, skin conductivity, and muscle shape and size [8]. Although the calibration of HGR classifiers for individuals can be a burden in real-world scenarios, it ironically emphasizes the potential of sEMG biometric systems to exploit the differences in these individual factors.

Several researchers have demonstrated the potential of sEMG biometric systems using a vector quantization and Gaussian mixture model [8, 9]. Subsequently, many researchers have deployed multichannel sEMG sensors to collect data in laboratories, followed by the implementation of various feature-extraction and classification algorithms. As reported in previous work [10], two-channel sEMG sensors were deployed, after which time-domain (TD) features were extracted to evaluate the user-identification performance of sensors using k-nearest neighbor (kNN), support vector machine, and artificial neural network models. Moreover, in previous work [11], a convolutional neural network (CNN)-based binary classifier was employed for user authentication using eight-channel sEMG sensors. Both studies [10, 11] reported accuracies of 81.6% and 95.0% using multichannel sEMG sensors, respectively. However, the small size of the dataset—which included fewer than 10 subjects—presented limitations.

To overcome these limitations, many studies [1, 12, 14, 15] have increased confidence in the experimental results by deploying larger datasets that collected eight-channel sEMG signals from more than 20 subjects. In previous work [1], sEMG signals were collected from 24 subjects performing 16 different gestures, followed by an assessment of their user-identification and verification performance. Additionally, the study obtained a 3.5% verification equal error rate (EER). Notably, the study introduced the concept of multifactor authentication based on gesture codes—which are combinations of multiple gestures, such as passwords—achieving a lower verification EER (1.1%). Although He and Jiang [1] employed a frequency-domain (FD) feature, a subsequent study deployed both TD and FD features and compared their performance [12]. Moreover, in previous work [13], a dataset comprising 56 subjects was collected, and false acceptance and rejection rates of 0.2% and 2.9%, respectively, were obtained using a kNN binary classifier. Reference [14] assessed the performance of sEMG-based biometric identification and verification using three datasets comprising 65 subjects. Notably, these studies were limited by several factors. First, they employed only traditional machine-learning algorithms, neglecting deep-learning (DL)-based methods, which have recorded breakthroughs in the fields of facial and speech recognition [24–27]. Second, they did not employ a test protocol that considered unknown identities, which is crucial for practical biometric authentication systems.

Lu and others [15] proposed a convolutional Siamese network [28] based on the wavelet transform features of sEMG signals. In this study, the model was trained using a contrastive loss function [28] with 4200 genuine and impostor pairs generated from 21 subjects, achieving 99.29% verification performance. Additionally, to evaluate the performance of this model against unknown subjects, new impostor pairs were generated from four new subjects, with a performance of 97.76% being recorded. Moreover, Fan and others [17] proposed a smartphone unlocking system based on sEMG signals, and a convolutional Siamese network was proposed. In this system, picking-up and putting-down gestures were collected from 40 subjects, achieving a verification accuracy of 92.06%. This study was the first to report the verification performance of unknown subjects under an open-set verification protocol [29].

Although user-independent models based on Siamese networks have been studied, the number of gestures has been limited, and detailed studies on improving the generalization performance of these models have been insufficient. In this study, we designed a robust user-independent model for sEMG-based biometric authentication and experimentally verified its performance. The proposed model was constructed based on a convolutional Siamese network architecture with an N-pair loss function [30]. The N-pair loss function could determine a better optimum than the contrastive loss function by simultaneously considering multiple negative examples. Second, we modified the input format and padding method of the model to exploit the shift-equivariant property of convolution operations. These modifications enabled the model to handle small shifts in the EMG channels, which could vary among users. Third, we applied two types of data-augmentation (DA) methods, time-series augmentation (TSAug) and spectrogram augmentation (SpecAug), to further improve the generalization performance.

The proposed model was trained and evaluated using a publicly available dataset, Gesture Recognition and Biometrics electroMyogram (GRABMyo), comprising 16 gestures collected from 43 subjects over 3 days [31]. To systematically evaluate the user-independent model, we established the following four test protocols by combining normal and leaked tests based on a gesture code scenario [1] with closed- and open-set tests based on the inclusion of unknown subjects: closed-normal (CN), closed-leaked (CL), open-normal (ON), and open-leaked (OL) protocols. These protocols were used to verify the effectiveness of each component of the model as well as the effectiveness of the deployed DA method. The final trained model obtained EERs of 2.58% and 3.68% for the CN and CL tests, respectively, and 5.62% and 8.94% for the ON and OL tests, respectively. Additionally, we conducted a multiday analysis to assess the robustness of the model to variations on different days. The results demonstrated improved performance in open-set tests, whereas the performance degraded in closed-set tests. The remainder of this paper is structured as follows. Section 2 describes the dataset employed for training and testing the model and presents the details of the proposed user-independent model, DA methods, and training and evaluation procedures. Section 3 presents and discusses the experimental results obtained using the four test protocols. Finally, Section 4 concludes the paper.

2 | MATERIALS AND METHODS

2.1 | Dataset

The publicly available forearm and wrist EMG dataset GRABMyo [31] were used to verify the performance of the proposed authentication model. The GRABMyo dataset comprised data from 43 subjects collected over three sessions on different days and 16 hand gestures with seven repetitions. Each signal was recorded using an EMGUSB2+ device (OT Bioelettronica, Italy) at a sampling rate of 2048 Hz. Moreover, the GRABMyo dataset provided sEMG signals measured using 16 monopolar electrodes comprising two forearm rings. We used eight-channel sEMG signals obtained from a forearm ring near the elbow. Sixteen gestures are depicted in Figure 1, namely, the lateral prehension (LP), thumb adduction (TA), thumb and little finger opposition (TLFO), thumb and index finger opposition (TIFO), thumb and index finger extension (TIFE), thumb and little finger extension (TLFE), index and middle finger extension (IMFE), little finger extension (LFE), index finger extension (IFE), thumb extension (TE), wrist extension (WE), wrist flexion (WF), forearm supination (FS), forearm pronation (FP), hand open (HO), and hand close (HC) gestures.

2.2 | Feature extraction

EMG-based gesture recognition and biometric studies using DL models use three types of input—that is, raw EMG signals, TD features, and FD features. Côté-Allard and others [32] demonstrated that the spectrogram extracted from raw signals improved the gesture recognition accuracy compared with raw signals. The spectrogram inherently reflects the temporal changes in the FD using the sliding-window technique. In our preliminary experiments, we compared Hudgin's TD features [21]—comprising the root mean error, zero crossing, slope-sign changes, and waveform length—with the spectrograms generated using an improved discrete Fourier transform (iDFT) [33]. We selected spectrogram features as inputs for our model because they showed better performance than the TD features. The iDFT features could be formally calculated as follows:

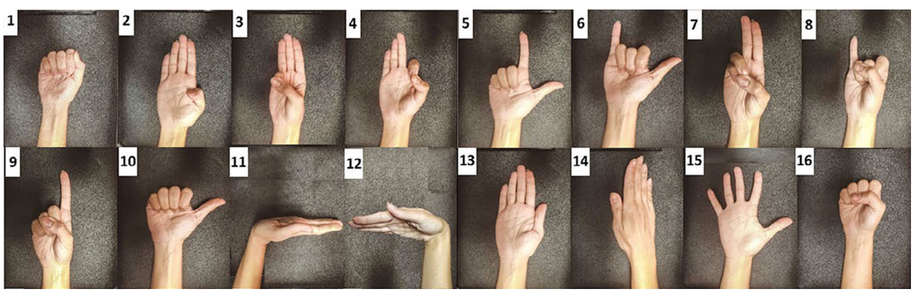

FIGURE 1 Sixteen gestures in the Gesture Recognition and Biometrics electroMyogram (GRABMyo) dataset: (1) lateral prehension (LP), (2) thumb adduction (TA), (3) thumb and little finger opposition (TLFO), (4) thumb and index finger opposition (TIFO), (5) thumb and index finger extension (TIFE), (6) thumb and little finger extension (TLFE), (7) index and middle finger extension (IMFE), (8) little finger extension (LFE), (9) index finger extension (IFE), (10) thumb extension (TE), (11) wrist extension (WE), (12) wrist flexion (WF), (13) forearm supination (FS), (14) forearm pronation (FP), (15) hand open (HO), and (16) hand close (HC).

$$\mathrm{DFT}_i = F\left[\frac{1}{n_i}\sum_{j=1}^{n_i} X\left(f_{i,j}\right)\right],\ i = 1,2,\ldots,6, \tag{1}$$

where the entire 20–450 Hz frequency band is evenly divided into six sub-bands, n_i denotes the length of the ith sub-band, and $f_{i,j}$ denotes the jth frequency within the ith sub-band. Further, $X(f)$ denotes the DFT at frequency f, and $F[\cdot]$ denotes a nonlinear transformation, which we set to the logarithm. Before applying (1), a sliding window was applied to the input EMG signals. The window length and stride were set to 200 ms and 100 ms, respectively.

2.3 | Convolutional Siamese network

A classification-based standard CNN generally requires a large volume of data for the effective training of a robust model. However, large datasets with hundreds of identities are not available for sEMG biometrics owing to the time-consuming nature of collecting sEMG signals. Siamese networks do not require large amounts of data and can handle unknown identities through similarity metric learning [28]. Consequently, we adopted a Siamese network as the user-independent DL model.

The proposed DL model was designed based on the Siamese network architecture (Figure 2). Although a basic Siamese network comprises two identical subnetworks sharing weights, the proposed method was expanded to contain $N+1$ subnetworks, as it uses an N-pair loss function. The N-pair loss function outperforms the contrastive loss function in computer vision applications. However, to the best of our knowledge, this has never been investigated in the context of sEMG biometric authentication. The N-pair loss function is described in the following subsections.

Moreover, the iDFT feature—represented by $x \in \mathbb{R}^{T\times F\times C}$—is the input of the proposed network (where T denotes the number of frames after applying the sliding window, F denotes the number of frequency sub-bands, and C denotes the number of sEMG channels). The duration of the input EMG signals can affect the verification performance, as longer durations yield improved accuracy [1]. Considering the trade-off between performance and usability, we set the input duration to 1.8 s, yielding the iDFT feature for 17 frames.

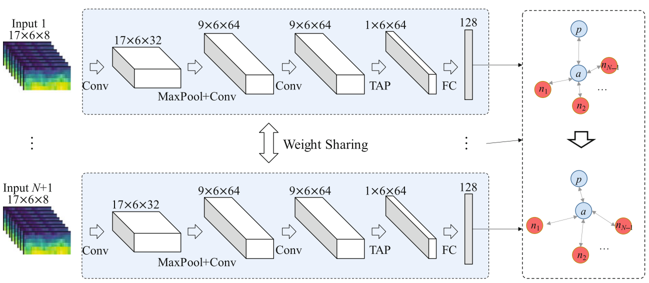

FIGURE 2 Proposed convolutional Siamese network with N-pair loss function.

The proposed network comprised three convolutional layers, two pooling layers, and a fully connected layer. First, input feature x was convolved using 32 convolutional filters with zero padding to output a $17 \times 6 \times 32$ feature map. Thereafter, the feature map was downsampled through a max-pooling layer to reduce the number of parameters and computations in subsequent layers. Next, two consecutive convolutional layers were applied, outputting a $9 \times 6 \times 64$ feature map that could be aggregated along the time axis using the statistical mean in the temporal average pooling layer to output a 6×64 feature map. Finally, we obtained a 128-dimensional embedding vector through a fully connected layer. All the convolutional layers were followed by batch normalization and rectified linear unit activation functions.

After designing this basic network—namely, the SiameseNetV1 model—we considered the robustness of the rotation of the channels in the model. Although sEMG authentication systems can have sensor placement guides in experimental environments, the positions of the electrodes may differ slightly for each subject. This is particularly true for armband-type sensors in which the electrodes are evenly spaced perpendicular to the forearm, allowing them to shift simultaneously.

To address this problem, we devised two simple, yet effective, methods for the convolution and padding operations. First, we changed the input format of the network from $T\times F\times C$ (TFC) to $T\times C\times F$ (TCF) such that the channel dimension preceded the frequency dimension. The two-dimensional (2D) convolution operation was shift equivariant with respect to the spatial dimensions corresponding to the first and second axes of the input tensor. Although the order of the frequency sub-bands of the input tensor remained constant regardless of the external environment, the sEMG channels shifted depending on the sensor placement. Consequently, applying the shift-equivariant property to the channel dimension allowed the convolution filter to share weights even when the channel shifted, thereby reducing the number of convolution filters. Assuming that the channel dimensions were not shift equivariant, additional convolutional filters corresponding to each shift had to be learned. Additionally, the padding method was changed from zero to circular to preserve the shift-equivariant property. Figure 3 shows that the convolution with zero padding disrupted the shift-equivariant property, where

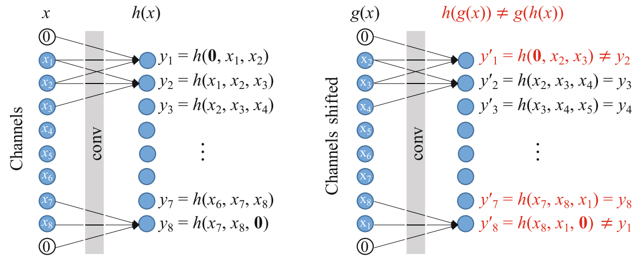

FIGURE 3 An example where the shift-equivariance property is disrupted in convolution operation with *zero padding*.

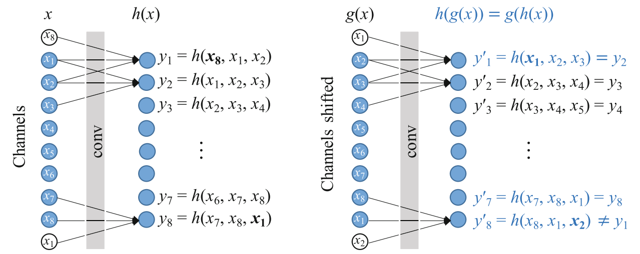

FIGURE 4 An example where the shift-equivariance property is preserved in the convolution operation with *circular padding*.

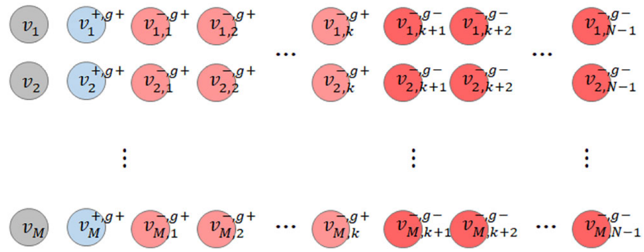

FIGURE 5 Example of batch constructions using gestures. The superscript (a, b) of v represents the positive $(+)$ or negative $(-)$ identifier: a and b represent the class and gesture, respectively.

$g(h[x]) \neq h(g[x])$, and $g(\cdot)$ and $h(\cdot)$ denote shift and convolution operations, respectively. Conversely, the convolution with circular padding maintained the shift-equivariant property, where $g(h[x]) = h(g[x])$ (Figure 4). Circular padding wrapped the input such that the values from the opposite edge could be used to pad the input. Circular padding was applied along the channel dimensions of the input, whereas the time dimension was padded to zero.

Based on these two methods, an optimized user-independent model was designed—that is, the SiameseNetV2 model—which was effectively adapted to small shifts in the sEMG channels across different users. The performance of the model was experimentally proven, as described in Section 3.

2.4 | Loss function

The proposed Siamese network was designed to learn embedding representations such that data from the same class could be pulled closer together, whereas those from different classes were pushed farther apart. Typical loss functions for Siamese networks—for example, the contrastive [28] and triplet [34] functions—have been widely employed. However, they suffer from problems such as their slow convergence and poor local optima owing to the use of only one negative example. To address these problems, an N-pair loss function that uses multiple negative examples was introduced [30]. An N-pair loss function was employed to train the proposed user-independent model, which can be expressed as follows:

$$L\left(\{x, x^+, \{x_i\}_{i=1}^{N-1}\}; \nu\right) = \log\left(1 + \sum_{i=1}^{N-1} \exp\left(\nu^{\mathrm{T}}\nu_i - \nu^{\mathrm{T}}\nu^+\right)\right), \tag{2}$$

where x denotes an anchor, x^+ denotes a positive example of x, and $\{x_i\}_{i=1}^{N-1}$ denote negative examples.

An embedding, denoted by ν, retains all the superscripts and subscripts of the corresponding example x. This loss function enforces the discriminative power between classes by jointly exploiting the similarities from one positive pair and $N-1$ negative pairs. Cosine similarity was employed, which considers only the angle between normalized embeddings. However, the inner product of the normalized embeddings was less than 1, which complicated the optimization. Thus, we introduced a regularization loss function for the unnormalized embedding vectors [30]. The N-pair loss function considers multiple negative examples; it must not consider hard negative class mining if the number of classes is smaller than N.

The GRABMyo dataset contains limited classes, allowing the inclusion of all negative classes in one tuplet. Regarding gestures, simply including all gestures in one tuplet may not be the optimal approach. He and Jiang [1] demonstrated that the verification performance for a normal test was higher than that for a leaked test, indicating a greater challenge in distinguishing between users performing the same gesture and those performing different gestures.

Based on previous studies, a new batch-construction method was proposed comprising the same (or positive) and different (or negative) gestures relative to the anchor (Figure 5). Experimental results of the proposed batch-construction strategy are presented in Section 3.

2.5 | Data augmentation

When analyzing sEMG signals, collecting data from numerous subjects can be challenging because of the accompanying inconvenience of tiredness and time-consuming repetitions. To address this problem, we applied two DA methods—that is, the TSAug [35] and SpecAug methods [36]. First, we applied the TSAug, Gaussian noise (GN), and magnitude warping (MW) methods to the sEMG signals. Both methods facilitated a performance improvement in previous work [35].

GN that emulates additive sensor noise can be achieved by adding a noise signal with a desired signal-to-noise ratio (SNR). The augmented signal (x^*) can then be calculated from the original signal (x) as follows:

$$\begin{gathered} x_i^* = x_i + n_i, \\ n_i \sim N\left(\mu = 0,\ \sigma = \sqrt{\frac{x_i^2}{\mathrm{SNR}}}\right), \end{gathered} \tag{3}$$

where i denotes the sample index, and n denotes the noise sampled from a Gaussian distribution with zero mean and variance equal to the ratio of the signal power to the SNR.

Conversely, the MW method deforms the signal amplitude by multiplying it with a randomly smooth curve. This smooth curve can be generated using cubic spline interpolation (*CubicSpline*) on a few random values, expressed as follows:

$$\begin{gathered} x_i^* = x_i \cdot s_i, \\ s = CubicSpline(r_1, \ldots, r_t), \\ r_j \sim N(\mu = 1,\ \sigma), \end{gathered} \tag{4}$$

Second, we applied SpecAug to the spectrogram, which is an input feature of the proposed model, and extracted it using the iDFT feature-extraction algorithm. In previous work [36], the SpecAug method exhibited performance improvements in speech recognition tasks by masking random parts of the time or frequency dimensions of the 2D spectrogram. However, the speech spectrogram used in this study exhibited two key differences. First, the sEMG spectrogram displayed a lower resolution than the speech spectrogram owing to its low sampling rate and limited frequency band. Second, a 3D spectrogram comprising multiple 2D spectrograms of multichannel sEMG signals should be considered.

Considering these differences, we made two modifications to the SpecAug method. First, we masked a spectrogram in only a single time frame or frequency band, as it comprised 17 time frames and six frequency sub-bands. Second, we repeated the spectrogram masking in some channel dimensions for a multichannel 3D spectrogram. Specifically, time masking (TM) was defined to ensure the masking of m_T time frames at c_T channels. Additionally, frequency masking (FM) was defined to ensure masking of the m_F frequency bands at c_F channels. The masking locations were randomly determined for each time, frequency, and channel dimension. Moreover, TM and FM could be applied simultaneously. Figure 6 shows examples of the augmented spectrograms.

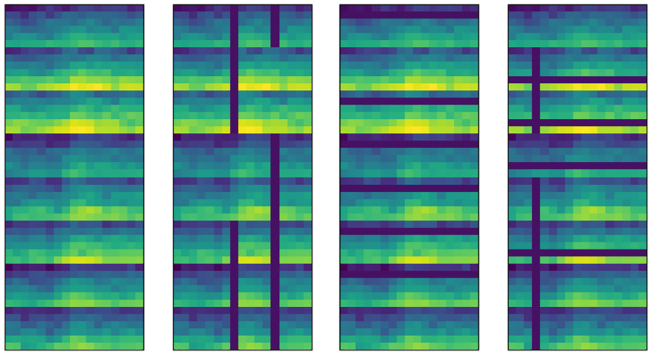

FIGURE 6 Examples of the application of the spectrogram augmentation (SpecAug) method to eight-channel sEMG signals. From left to right, the figure depicts the spectrogram without augmentation, time masking (TM), frequency masking (FM), and a combination of TM and FM. The eight spectrograms with a $T \times F$ shape are concatenated vertically for visualization.

2.6 | Training and evaluation

During training, the network was optimized using the ADAM optimizer at a learning rate of 0.001. Moreover, 7500 training iterations were applied with a batch size of 64, and the learning rate decreased by 1/10 after 5000 steps. The training and evaluation codes were implemented using the Python TensorFlow library.

To evaluate the proposed user-independent model, closed- and open-set test protocols were employed, which were distinguished by the inclusion of unknown subjects in the test dataset or lack thereof [29]. In the closed-set test—which typically includes only known subjects in the training dataset—the protocol was modified by introducing unknown subjects into the impostor pairs, thereby simulating attempts made by unknown users to access the system. In the open-set test, genuine and impostor pairs were generated using only unknown subjects, thereby allowing us to evaluate the generalization performance of the model.

To simultaneously conduct model training as well as the closed- and open-set tests, the dataset was divided as depicted in Figure 7.

First, the dataset was divided into two datasets based on the subjects, yielding $\text{train}_{\text{subj}}$ and $\text{test}_{\text{subj}}$ datasets. Second, for each fold, the data were divided into two subsets in a 5:2 ratio based on the trials, yielding $\text{train}_{\text{trial}}$ and $\text{test}_{\text{trial}}$ datasets. Finally, the following four subsets were obtained:

- $\textit{train} = \text{train}_{\text{subj}} - \text{test}_{\text{trial}}$
- $\text{test}_{\text{c}} = \text{test}_{\text{trial}} - \text{test}_{\text{subj}}$
- $\text{test}_{\text{cu}} = \text{test}_{\text{trial}} \cap \text{test}_{\text{subj}}$
- $\text{test}_{\text{o}} = \text{test}_{\text{subj}}$

where *train* denotes the training dataset, test_{c} denotes the closed-set test dataset with known subjects, test_{cu} denotes the closed-set test dataset with unknown subjects, and test_{o} denotes the open-set test dataset. Moreover, twofold cross-validation (CV) was applied to address the problem of insufficient data.

sEMG biometrics can be used to easily construct a multifactor authentication system by requiring a specific gesture code from a user. Assuming this system, three types of test protocols have been proposed—that is, the *normal*, *leaked*, and *self-test* protocols [1]. Here, a *normal* test was adopted in which the gesture code was known only to the legitimate user, as well as a *leaked* test in which the gesture code was leaked to the attacker.

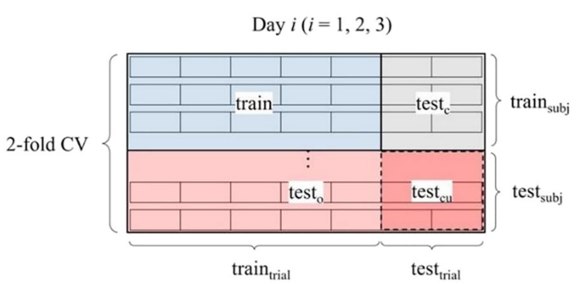

FIGURE 7 Data partitioning for within-day analysis, enabling simultaneous closed-set and open-set tests.

By combining both test types, four verification protocols could be established—that is, the CN, CL, ON, and OL verification protocols. Each generated verification pair is detailed in Table 1. Each trial lasted 5 s, and because the proposed model permitted 1.8 s of input, the test examples were concatenated and augmented by applying a sliding window with half-overlapping windows. In the GRABMyo dataset, 10 and 37 test examples per subject were used for the closed- and open-set tests, respectively.

For the closed-set test, 945 genuine pairs comprising combinations of 10 examples per subject were employed. The genuine pairs were the same in the *normal* and *leaked* tests; however, the impostor pairs differed. Regarding the impostor pairs for the CN test, 210 pairs were first generated between known subjects in test_{c}, and 462 pairs between known subjects in test_{c} and unknown subjects in test_{cu}. As each pair comprised different gestures, 10 752 pairs were obtained. Finally, the pairs were repeated for all examples, yielding 107 520 pairs. This could be expressed mathematically as follows:

$$\left(\binom{N_s^{\text{test}_c}}{2} + N_s^{\text{test}_c} \times N_s^{\text{test}_{cu}}\right) \times N_g \times N_e, \tag{5}$$

where N_s^{test}, N_g, and N_e denote the numbers of subjects for a test subset, gestures, and examples, respectively. For the CL test, 6720 impostor pairs were generated using the same procedure, incorporating the constraint that the gestures within each pair must be identical, as required by the *leaked* test.

For the open-set test, 14 652 genuine pairs, comprising a combination of 37 examples per subject, were used. To generate the impostor pairs, two different subjects were selected as pairs from the 22 subjects in the $\textit{test}_o$ subset. In the *normal* test, the gestures of each pair differed, generating 3696 pairs. When this process was

TABLE 1 Overview of the verification pairs across the four tests on the Gesture Recognition and Biometrics electroMyogram (GRABMyo) dataset.

	Closed set		Open set	
	Normal	**Leaked**	**Normal**	**Leaked**
Number of Subjects	43 (21 + 22)		22	
Number of Examples	10		37	
Number of Genuine Pairs	945	945	14 652	14 652
Number of Impostor Pairs	107 520	6720	136 752	8547

repeated for all examples, 136 752 pairs were generated. Mathematically, this can be expressed as follows:

$$\binom{N_s^{\text{test}_o}}{2} \times N_g \times N_e. \tag{6}$$

Regarding the OL test, only 8547 impostor pairs were used because of the constraint that the gestures of both pairs had to be the same.

For more practical implementation in real-world scenarios, the intersession variability of a user should be evaluated in addition to the within-session performance. Inspired by Pradhan and others [18], a multiday analysis was conducted, which included within-day (WD), single-cross-day (SCD), and cumulative cross-day (CCD) analyses. WD analysis served as the primary evaluation method for the detailed performance of the proposed methods, including the model design, loss function, and DA. It used the training and test datasets from the same day. By contrast, the SCD and CCD analyses were conducted using datasets from different days, as illustrated in Figure 8. The SCD analysis used 1 day as the training dataset and the remaining days as the test dataset. The training dataset in the SCD analysis was the same as that used in the WD analysis. The CCD analysis used 2 days as the training dataset and another day as the test dataset, which enhanced the robustness of the intersession variability. For each analysis, the average performance over 3 days was reported.

3 | RESULTS AND DISCUSSIONS

A robust user-independent model for sEMG biometric authentication was designed based on three main features—that is, an *N*-pair loss function, a channel shift-equivariant model, and DA. In this section, we evaluate the verification performance of the proposed model based on the EERs calculated from the receiver operating characteristic curve, which plots the true and false positive rates across a range of thresholds. Additionally, multiple EERs were obtained after the verification protocol for the CV and multiple gestures. We employed the mean values of the EERs to compare their performances.

3.1 | Loss function

Table 2 presents the results of comparing the performances of the proposed SiameseNetV1 model based on the two loss functions (contrastive and *N*-pair loss functions). The *N*-pair loss function resulted in an overall decrease in the EER compared with the contrastive loss function using hard negative mining (HNM). In the CN

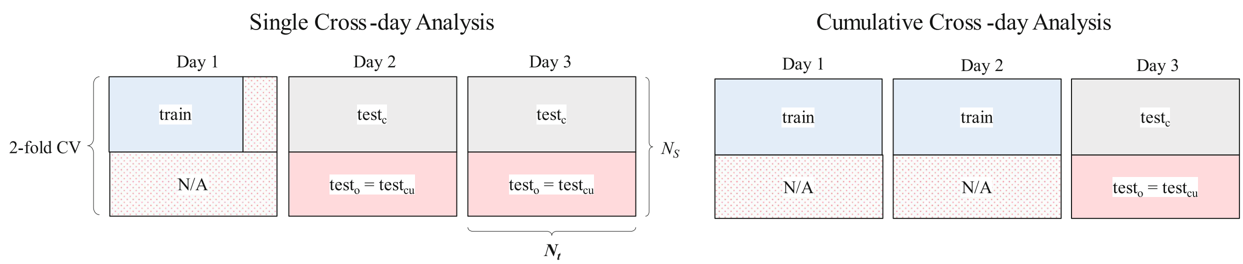

FIGURE 8 Visualization of single cross-day and cumulative cross-day analyses, with data partitioning for simultaneous closed-set and open-set tests.

TABLE 2 Performance comparison of two loss functions with different batch-construction methods.

		Closed set		Open set	
Loss function	Batch construction	Normal	Leaked	Normal	Leaked
Contrastive	HNM	3.50	4.78	8.15	**10.33**
N-pair	SG	4.37	**4.65**	8.42	10.60
N-pair	RG	**2.97**	4.70	**6.57**	11.77
N-pair	HG11	3.11	4.70	6.61	11.20
N-pair	HG21	3.20	4.70	6.61	10.74

Abbreviations: HNM, hard negative mining; RG, random gesture; SG, same gesture.

and CL tests, the contrastive loss achieved EERs of 3.50% and 4.78%, respectively, with an average value of 4.14%. In the open-set tests, EERs of 8.15% and 10.33% were achieved, with an average value of 9.24%. This represents an increase of 4.8% in the average EER compared to the *N*-pair loss with the HG21 batch-construction method in the closed-set test and an increase of 6.5% in the open-set test. This demonstrates that the *N*-pair loss function outperformed the contrastive loss function in the EMG-based authentication and vision tasks.

Moreover, we experimented with three batch-construction methods for the *N*-pair loss function that considered the following gestures—that is, the same gestures (SGs), which generate all negative examples of a tuplet from the same gesture; random gestures (RGs), which randomly select gestures; and hybrid gestures (HGs), which combine SGs and RGs. The HG method was tested using different SG:RG ratios, specifically 1:1 (HG11) and 2:1 (HG21). As expected, the RG method achieved the lowest EERs in the *normal* test but the highest EERs in the *leaked* test. Similarly, the SG method exhibited the lowest EERs in the *leaked* test but the highest EERs in the *normal* test. Although the HG method did not achieve the best performance, it demonstrated reliable performance across both test types. In terms of the average EER, the closed-set test results were 4.51%, 3.84%, 3.91%, and 3.95% for the SG, RG, HG11, and HG21, respectively. For the open-set test, the EERs were 9.51%, 9.17%, 8.91%, and 8.68%, indicating that the HG21 method outperformed the others as a superior batch-construction method. Notably, the HG strategy displayed a performance trade-off between the *normal* and *leaked* tests, depending on the combination ratio of the SG and RG.

3.2 | Channel shift-equivariant model

Here, we evaluated the effectiveness of the designed shift-equivariant channel in the SiameseNetV2 model for dealing with small shifts in the EMG channels across different users. The models deployed in this experiment were trained using the *N*-pair loss function and HG21 batch-construction method, the results of which are presented in Table 3. The first element of the SiameseNetV2 model involves changing the input format from TFC to TCF to incorporate shift equivariance in the convolution operations along the channel dimension instead of the frequency dimension. Consequently, an overall decrease was evident in the verification EERs for the TCF format compared to the TFC format. In particular, in the OL test, the TCF-trained model achieved an EER of 9.19%, which was 14.4% lower than that of the TFC-trained model.

The second element of the SiameseNetV2 model involves the application of circular padding to the convolution operations. As was evident, the application of circular padding yielded EERs of 2.78% and 3.95% for the CN and CL tests, respectively, and 5.68% and 9.01% for

TABLE 3 Performance comparison of the two design factors in the SiameseNetV2 model: the input format and padding method.

		Closed set		Open set	
Input format	Padding	Normal	Leaked	Normal	Leaked
TFC	Zero padding	3.20	4.70	6.61	10.74
TCF	Zero padding	**2.78**	4.05	5.72	9.19
TCF	Circular padding	**2.78**	**3.95**	**5.68**	**9.01**
TCF	No-padding	3.12	4.93	6.84	11.92

TABLE 4 Performance comparison of the time-series augmentation (TSAug) method, which includes Gaussian noise (GN) and magnitude warping (MW), with different hyperparameters.

		Closed set		Open set	
Method	Hyperparameter	Normal	Leaked	Normal	Leaked
-	-	2.78	3.95	5.68	9.01
GN	SNR = 30	2.62	3.85	5.72	9.06
GN	SNR = 35	**2.59**	3.79	5.72	8.99
MW	$\sigma = 0.1$	2.64	3.82	5.75	9.03
MW	$\sigma = 0.2$	2.65	**3.75**	**5.67**	**8.91**

the ON and OL tests, respectively. These EERs were lower than those obtained by applying zero padding, demonstrating that the maintenance of shift equivariance in the channel dimension was a crucial property of the sEMG-based authentication model. We also conducted experiments without padding, which yielded higher EERs than the other two padding methods. This was because the no-padding method could not maintain the shift-equivariant property or reduce the channel dimensions of intermediate feature maps, thereby limiting the feature space.

3.3 | Data augmentation

DA experiments were conducted using the SiameseNetV2 model, which demonstrated the best performance, as reported in the previous subsection. Table 4 presents the experimental results of applying the TSAug method to the EMG signals. Both the GN and MW methods contributed to improved performance in the closed-set tests, regardless of the hyperparameters. The GN method improved the performance by approximately 6.8% in the CN test and 4.0% in the CL test. Similarly, the MW method achieved improvements of 4.7% and 5.0%, respectively. By contrast, for the open-set tests, both the GN and MW methods showed no major performance improvement or only marginal gains. This can be interpreted as the TSAug method primarily increasing intraclass variation without greatly affecting interclass variation.

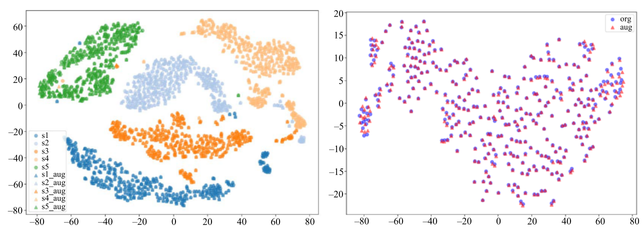

FIGURE 9 Visualization of the t-SNE distribution of the original (circles) and Gaussian noise (GN)-augmented data (triangles). The left panel illustrates five subjects, represented in different colors, whereas the right panel focuses on a single subject.

Figure 9 shows the t-SNE distributions of the original and augmented data, providing a visualization of these variations. In the figure with multiple subjects, the augmented data appear to align closely with the distribution of the original data. However, a closer examination of a single subject reveals that new samples were randomly shifted within the original data, leading to an increase in intraclass variation.

Interestingly, a negligible performance gap was evident between the GN and MW methods, even with varying hyperparameters such as the SNR or sigma. These results contrast with those of previous work [35], which showed considerable differences with respect to the parameters. This discrepancy is likely because of differences in the target tasks and datasets.

Table 5 lists the experimental results obtained using the SpecAug method. The TM method facilitated performance improvements in the closed-set tests but showed minimal improvement in the open-set tests compared with no augmentation, which was consistent with the results of the TSAug method. For the c_T hyperparameter, the best performance was achieved at $c_T = 2$; however, there was a small performance gap within the range of 2–8. This suggests that the TM method of creating only a subset of channels in multichannel 3D spectrograms had limited effectiveness. Conversely, in the FM method, the performance decreased compared to that of the baseline model across all hyperparameters. This could be attributed to the fact that the frequency sub-bands of the

TABLE 5 Performance comparison of the spectrogram augmentation (SpecAug) method, which includes time masking (TM) and frequency masking (FM), with different hyperparameters.

		Closed set		Open set	
Method	Hyperparams.	Normal	Leaked	Normal	Leaked
-	-	2.78	3.95	5.68	9.01
TM	$m_T = 1, c_T = 8$	2.63	3.75	**5.66**	8.95
TM	$m_T = 1, c_T = 6$	**2.57**	3.71	5.72	9.01
TM	$m_T = 1, c_T = 4$	2.64	3.75	5.70	8.96
TM	$m_T = 1, c_T = 2$	2.61	**3.66**	5.68	**8.93**
FM	$m_F = 1, c_F = 8$	2.91	4.47	5.95	9.78
FM	$m_F = 1, c_F = 6$	2.88	4.54	5.75	9.94
FM	$m_F = 1, c_F = 4$	2.96	4.53	5.84	9.93
FM	$m_F = 1, c_F = 2$	2.87	4.32	5.83	9.80

TABLE 6 Performance comparison of the single and mixed data-augmentation (DA) methods.

Method	Hyperparams.	Closed set		Open set	
		Normal	Leaked	Normal	Leaked
-	-	2.78	3.95	5.68	9.01
MW	$\sigma = 0.2$	2.65	3.75	5.67	**8.91**
TM	$m_F = 1$, $c_F = 2$	2.61	**3.66**	5.68	8.93
MW + TM	$\sigma = 0.1$, $m_F = 1$, $c_F = 6$	**2.58**	3.68	**5.62**	8.94

Abbreviations: MW, magnitude warping; TM, time masking.

TABLE 7 Ablation study of the proposed methods in terms of the channel shift-equivariant model, loss function, and data-augmentation (DA).

Methods			Closed set		Open set	
Model	Loss	DA	Normal	Leaked	Normal	Leaked
SiameseNetV1	Contrastive	N	3.50	4.78	8.15	10.33
SiameseNetV1	*N*-pair	N	3.20	4.70	6.61	10.74
SiameseNetV2	*N*-pair	N	2.78	3.95	5.68	9.01
SiameseNetV2	*N*-pair	Y	**2.58**	**3.68**	**5.62**	**8.94**

spectrogram were independent and did not overlap in the frequency interval, unlike in the time frames. Additionally, the frequency resolution was low, indicating that each frequency sub-band played a crucial role in the model performance.

Additionally, we experimented using a combination of the MW and TM (MW + TM) methods, which were effective for the previous TSAug and SpecAug methods. As shown in Table 6, the MW + TM method yielded mean EERs of 2.58% and 3.68% for the closed-set tests and 5.62% and 8.94% for the open-set tests. Although we anticipated that combining different types of augmentation could enhance performance, the results did not support this hypothesis. However, it cannot be definitively concluded that the hybrid augmentation method was ineffective, as a performance improvement was evident in the cross-day analysis. Further details are provided in Section 3.5.

3.4 | Final user-independent model

Here, we summarize the experimental results of the proposed methods in terms of the model design, loss function, and DA (Table 7). The baseline model, trained using the SiameseNetV1 model and contrastive loss function, exhibited mean EERs of 3.50%, 4.78%, 8.15%, and 10.33% for the CN, CL, ON, and OL tests, respectively. When the first method—that is, the *N*-pair loss function—was applied to the baseline model, the EERs decreased by 8.6%, 1.7%, and 18.9% for the CN, CL, and ON tests, respectively, but increased by 4.0% for the OL test, compared with the baseline model. The channel shift-equivariant SiameseNetV2 model demonstrated further performance improvements compared to the first method, obtaining mean EERs of 13.1%, 16.0%, 14.1%, and 16.1% for the CN, CL, ON, and OL tests, respectively. By applying the DA method to the SiameseNetV2 model, further EER reductions of 7.2%, 6.8%, 1.1%, and 0.8% for the CN, CL, ON, and OL tests, respectively, were achieved. The *N*-pair loss function contributed to the overall improvement in model performance by obtaining a better optimum. However, its contribution to the generalization performance may be controversial, as it resulted in a performance reduction in the OL test. Concurrently, the channel-shift-equivariant design played a crucial role in improving the generalization performance of the user-independent model by demonstrating consistent performance improvements in the closed- and open-set tests. Similarly, the DA method improved the generalization performance of the model, although its impact was moderate. Notably, the final model trained using these three methods obtained EERs of 2.58%, 3.68%, 5.62%, and 8.94% for the CN, CL, ON, and OL tests, respectively.

The performances of each gesture in the final model are shown in Figure 10 and Figure 11. A ranking analysis of the results from the *normal* and *leaked* tests revealed common patterns among the gestures.

For the closed-set test, the TLFE, TIFO, and WE gestures consistently appeared in the top five with low EERs,

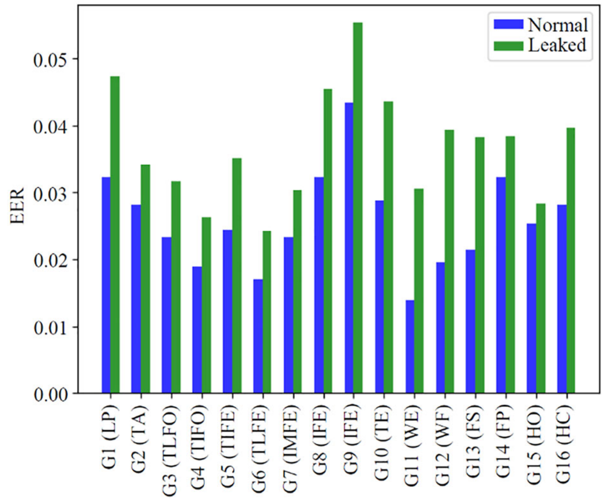

FIGURE 10 Closed-set verification performances of the gestures for the *normal* and *leaked* tests.

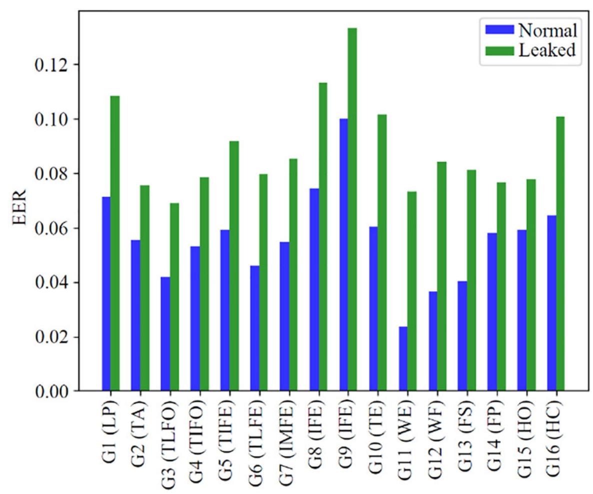

FIGURE 11 Open-set verification performances of the gestures for the *normal* and *leaked* tests.

whereas the IFE, LFE, LP, and TE gestures were included in the top five with high EERs. Comparing the two groups, gestures involving the movement of two fingers or the wrist demonstrated better performance than those involving the movement of a single finger. This suggests that gestures that induced greater muscle contractions were likely to result in higher performance. Additionally, in the *normal* test, the performance of the WE and WF gestures exhibited considerable improvement compared to the *leaked* test. This could be attributed to the fact that the *normal* test reflected not only intersubject differences caused by physiological characteristics but also inter-gesture differences. According to previous studies [37, 38], the WE and WF gestures showed higher recognition performance than other gestures. This could have contributed to the improved results evident in the *normal* tests. In the open-set test, the rankings of some gestures changed; however, the results were largely consistent with those of the closed-set test.

3.5 | Multiday analysis

Table 8 presents the results of the multiday analyses. For both the *normal* and *leaked* tests in the closed set, the cross-day analyses exhibited a considerable drop in performance compared to the WD analysis. By contrast, in the open-set tests, the cross-day analyses performed better than the WD analysis. In particular, the SCD analysis—which trained a model in one session and tested it in different sessions—yielded an average EER that was 1.8 times higher for the closed-set tests and 12.4% lower for the open-set tests compared to the WD analysis. This indicates that the model trained using a single session had limitations in learning nonstationary factors from different sessions while demonstrating the generalization capability to unknown subjects. The CCD analysis revealed approximately 20% lower EERs than

TABLE 8 Performance comparison of the three multiday analyses.

		Closed set		Open set	
Method	Multiday	Normal	Leaked	Normal	Leaked
Pradhan et al. [18]	WD	**2.80**	**4.00**	-	-
	SCD	13.30	18.10	-	-
	CCD	9.00	14.50	-	-
SiameseNetV2 (Ours)	WD	**2.78**	**3.95**	5.68	9.01
	SCD	4.75	7.24	5.09	7.78
	CCD	3.77	5.60	**4.03**	**6.18**

Abbreviations: CCD, cumulative cross-day; SCD, single-cross-day; WD, within-day.

TABLE 9 Performance comparison of the single and mixed data-augmentation methods for cross-day analyses.

		Closed set		Open set	
Multiday	**DA**	**Normal**	**Leaked**	**Normal**	**Leaked**
SCD	-	4.75	7.24	5.09	7.78
	MW	4.71	7.14	5.05	7.72
	TM	4.73	7.13	5.03	7.66
	MW + TM	**4.64**	**7.08**	**4.92**	**7.56**
CCD	-	3.77	5.60	4.03	6.18
	MW	3.78	5.62	4.04	6.20
	TM	**3.74**	**5.55**	4.01	6.12
	MW + TM	**3.74**	5.58	**3.97**	**6.11**

Abbreviations: CCD, cumulative cross-day; DA, data-augmentation; MW, magnitude warping; SCD, single-cross-day; TM, time masking.

the SCD analysis for both the closed- and open-set tests. These results were consistent with the findings of Pradhan and others [18], suggesting that training with multisession data could effectively handle intersession variability.

Compared with Pradhan and others [18], the proposed method achieved approximately 60% lower EERs for both the SCD and CCD analyses. Notably, Pradhan and others [18] were the first to introduce multiday analyses, including the WD, SCD, and CCD analyses, on the GRABMyo dataset, and provide a baseline performance using a Mahalanobis distance-based method. This highlights that the proposed DL model was more robust to intersession variability than their method. DA is another potential method for addressing session variability. As shown in Table 9, applying DA to SCD analysis led to noticeable performance improvements, particularly when the two augmentation methods were combined. Conversely, in the CCD analysis, the effect of DA was minimal. This suggests that the effectiveness of the proposed DA methods could be limited when applied to multisession datasets.

4 | CONCLUSIONS

In this study, we designed and evaluated a user-independent model for sEMG-based biometric authentication by using a convolutional Siamese network. To ensure the robustness of the model for different users, we designed a user-independent model based on three methods—that is, the channel-shift-equivariant design, *N*-pair loss function, and DA methods. We also developed four test protocols—namely, the CN, CL, ON, and OL protocols—for the GRABMyo dataset by combining closed- and open-set scenarios with a gesture code scenario. The effectiveness of the three methods deployed in the proposed model was assessed based on four test protocols. The final model achieved mean EERs of 5.62% and 8.94% for unknown subjects with preserved and leaked gesture codes, respectively. In a multiday analysis, the proposed model demonstrated robustness to intersession variability, achieving 60% lower EERs compared to previous works [18].

In the future, optimization studies should be conducted to further enhance the performance of the model. Previous research [32] compared gesture recognition performance using raw signals, spectrograms, and continuous wavelet transform features as inputs for CNN models. Similarly, Pradhan and others [12] investigated the performance of distance-based authentication methods using TD and FD features. Based on these studies, further research to optimize the performance of the Siamese-network-based user authentication model by exploring various combinations of input features and DL architectures—such as CNNs and recurrent neural networks—would be meaningful.

CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest.

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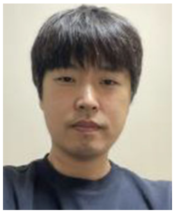

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